Algorithm 4 Random walk

function RANDOMWALK($q_{\text{start}}, q_{\text{goal}}, \Delta$)
    $q \leftarrow q_{\text{start}}$; $P \leftarrow (q_{\text{start}})$
    repeat
        $\theta \leftarrow$ RANDOMANGLE
        $q_{\text{new}} \leftarrow q + \Delta\,(\cos\theta, \sin\theta)$
        if $\neg$COLLISIONFREE(q, q_{new}) **then**
            continue
        end if
        $q \leftarrow q_{\text{new}}$; APPEND($P, q$)
        if $\|q - q_{\text{goal}}\| \leq \Delta$ and COLLISIONFREE(q, q_{goal}) **then**
            return APPEND(P, q_{goal})
        end if
    until false
end function